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In January 2005, we touched upon and compared AGP cards. Our prediction that PCIe video cards would take some time to pick up in the market has been correct to an extent. However, the lower end of the spectrum has not conformed to this, and it's certain that the popularity of low-end PCIe cards is on the rise—while the mid-range and high-end segments have yet to make its mark.

In this comparison, we opted for the same method of categorising our cards—in three segments; low-end, mid-range and high-end, like we did last time. Most of the cards we received this time are also available in the AGP flavour except for the nVidia 5XXX series and the ATi 9XXX series.

Do refer to the AGP comparison in our January issue. This will help you gain an overall perspective into the performance difference between the two platforms.

We tested 32 cards this time in sum across all categories with a modified test process (See box *How we tested*). The test process was modified keeping in mind the powerful GPUs that are available on current video cards. We got to lay our hands on some of the best cards available in the market—and some not-so-available ones—including the GeForce 7800GTX and the 6800 Ultra Dual GPU.

LOW-END CARDS

Would you have imagined six months ago that PCI-Express products would be available for the low-end market so soon? PCI-Express has penetrated the budget segment big-time. This could partly be credited to the plummeting prices of PCI-Express motherboards which, as you're probably aware, we reviewed last month.

The eight PCI-Express low-end graphics cards we received were from Gainward, Gigabyte, MaxForce, MSI, PowerColor and XFX. Even the big players, thus, have a presence in this segment. The cards in this segment were obviously based on the lower-performing GPUs—the nVidia GeForce 6200TC and the ATi X300.

Features

Both the nVidia and ATi GPUs in this segment featured new memory management techniques. While nVidia featured TurboCache, ATi featured HyperMemory. (More on these technologies later). The Gainward, PowerColor and XFX GeForce 6200TC supported 256 MB, while the rest supported 128 MB of memory. Of these, only the PowerColor X300 256 MB card sported a 128-bit memory interface, while the rest had either a 64- or a 32-bit memory interface. We therefore expected the PowerColor X300 to be the top performing card in this category.

Most of the cards came bundled with a S-Video cable—except the MSI GeForce 6200TC 128 MB, which came only with a driver CD. Some cards came bundled with a DVI to VGA adapter. Dual-monitor, which was a feature found only on high-end cards just a few years ago, was avail-

PCI-Express cards are much faster than their AGP cousins, and once you start playing games on them, it shows

able on all the cards here—except for the Gigabyte GeForce 6200.

All the nVidia-based cards supported DirectX 9.0c, which essentially means Shader Model 3.0, whereas the ATi-based cards supported DirectX 9.0b, which means Shader Model 2.0. The core of the nVidia cards is clocked at 350 MHz, whereas that of the PowerColor X300 is clocked at 400 MHz—while the other ATi-based cards had a stock core speed of 325 MHz. The 256 MB nVidia cards had memories running at effective speeds of 700 MHz, while all the ATi based cards had their memories clocked at 400 MHz.

The GeForce 6200TC has all the features of the high-end cards from nVidia. So why are these cards in this category? It is crippled by only four pipelines and a single texture mapped unit per pipeline. Also, the core and memory clock speeds are much lower than the GT cards in the mid-range and the high-end. This, in addition to the shared memory, causes them to yield much lower performance than the mid-range and high-end cards. This is also applicable to the cards based on the ATi chips.

Performance

As expected, the top performer in this category was the PowerColor X300 256 MB card, with its comparatively higher clock speeds and a wider, 128-bit memory interface.

At 1024 x 768 in *Doom 3*, this card posted a score of 22.20 fps, followed closely by the Gainward GeForce 6200TC 256 MB 20.30. In *Half-Life 2* at 1024 x 768, it logged 65.07, while the Gainward scored 48.13 fps. The same was the case with *FarCry*—the PowerColor churned out 72.19 fps, while the Gainward followed with 58.80.

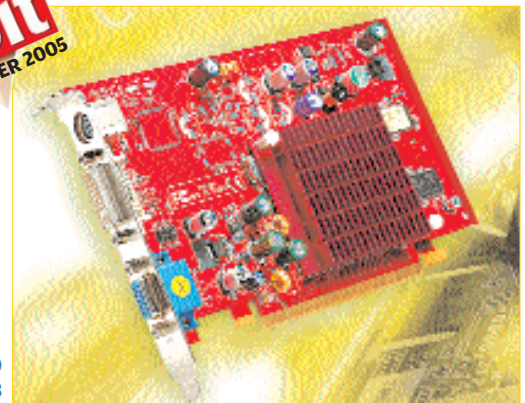
In 3DMark 03 at default settings, this card scored 3098, which is the highest in this category, followed by the Gainward at 2659.

Conclusion

Remember that this category, which we have called 'low-end', is nothing but the budget category, and pricing a card too high in this category is alienating the card from the consumers in this segment. The fight would have been between the MSI GeForce 6200 256 MB and the Gainward GeForce 6200 256 MB. But the MSI was priced way too high at Rs 6,300, whereas the Gainward was priced at Rs 3,950 - and thus, won the *Digit Best Buy Gold* because of its best price-to-performance ratio.

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Gainward GeForce 6200
256 MB



Photograph: Sandeep Patti Imaging Studio 5